

Week 9 Progress Summary

- Implemented two non-disruptive Data Science technical contributions in the Node backend: GET /api/data-profile and POST /api/top-correlated-pair.
- Added profiling logic to summarize dataset quality (row count, stream count, missing percentage, min/max/mean, and time range) for analytics readiness.
- Added Pearson-correlation-based pair selection logic to return the strongest correlated stream pair, sample size, and interpretation label (strong/moderate/weak).
- Integrated new capabilities through existing architecture layers (routes -> controllers -> services) to preserve current project structure and avoid breaking existing endpoints.
- Extended smoke tests to cover both new endpoints (success and validation paths), with full backend suite passing (7/7).
- Updated LLD and HLD documentation to reflect implemented /health, profiling, and correlation contributions, and regenerated Word outputs.
- Committed and pushed Week 9 implementation and documentation updates to the project branch for versioned traceability.
- Continued software architecture upskilling/certification and aligned new contribution work with architecture-first design and evidence standards.

SFIA Alignment (Week 9)

- Data analytics (DAAN): Implemented GET /api/data-profile and POST /api/top-correlated-pair to generate actionable data-quality and correlation insights from IoT streams.
- Programming/software development (PROG): Added backend service, controller, and route logic for new DS endpoints within the existing Node architecture.
- Functional testing (TEST): Extended smoke tests for new APIs (success + validation scenarios) and verified stable behavior with full test pass.
- Systems integration and build (SINT): Integrated new capabilities through existing layered flow (routes -> controllers -> services) without disrupting current runtime paths.
- Information security (SCTY): Preserved secure input handling by validating request payloads and returning controlled error responses for invalid stream selections.

- Knowledge management (KNOW): Updated HLD/LLD and evidence artifacts to keep implementation, architecture decisions, and traceability aligned for assessor review.

Week 8 Technical Snippets

Technical Contribution Evidence (Code Snippets)

1) Stream and filter Endpoint (Non-Disruptive Additive Control)

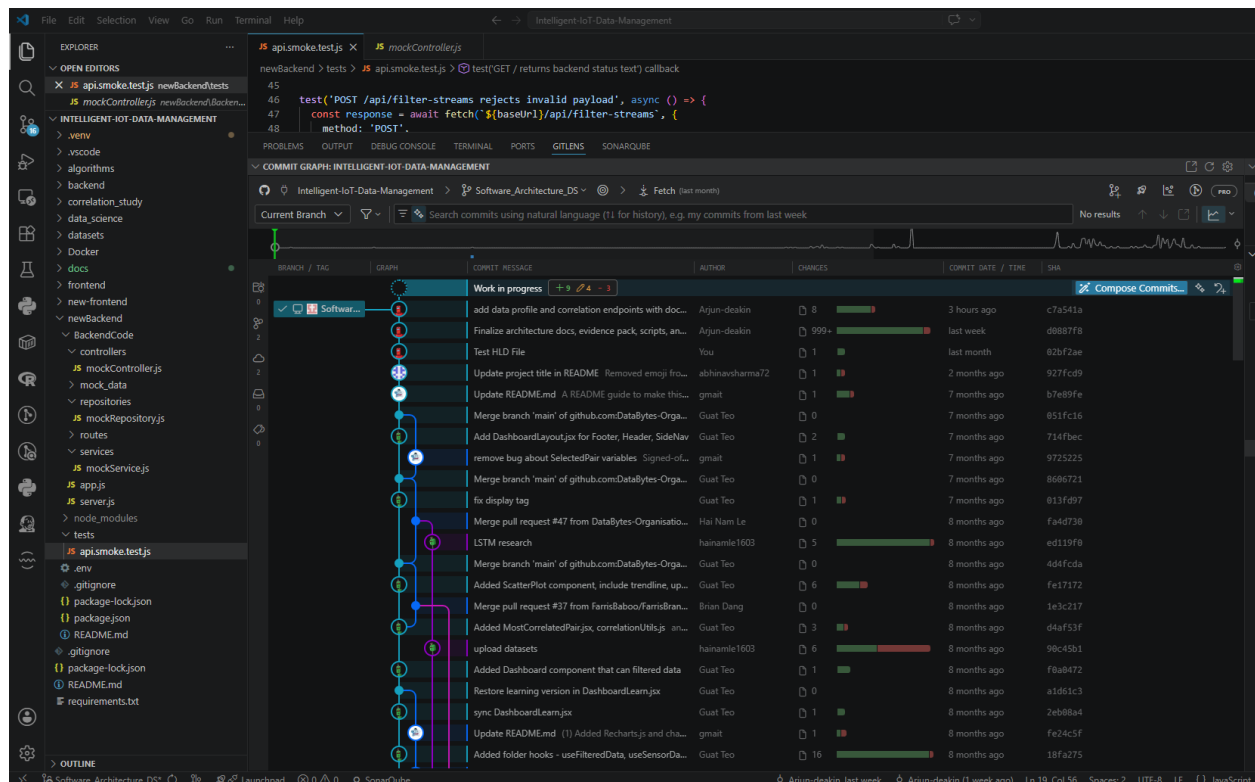
```

1 //handles the logic for processing mock data, using the repository for data access
2
3 const MockRepository = require('../repositories/mockRepository');
4 const mockRepository = new MockRepository();
5
6 //get all entries from the .json file
7 const readProcessedData = () => {
8   return mockRepository.getMockData();
9 };
10
11 const getAvailableStreamNames = () => {
12   const entries = mockRepository.getMockData();
13   if (!entries || entries.length === 0) return [];
14
15   const excludedKeys = ["created_at", "entry_id", "was_interpolated"];
16   return Object.keys(entries[0]).filter(key => !excludedKeys.includes(key));
17 };
18
19 const filterEntriesByStreamNames = (streamNames) => {
20   const entries = mockRepository.getMockData();
21
22   return entries.map(entry => {
23     const filteredEntry = {
24       created_at: entry.created_at,
25       entry_id: entry.entry_id
26     };
27
28     streamNames.forEach(name => {
29       if (entry[name] !== undefined) {
30         filteredEntry[name] = entry[name];
31       }
32     });
33
34     return filteredEntry;
35   });
36 };
37
38 const toNumber = (value) => {
39   const num = Number(value);
40   return Number.isFinite(num) ? num : null;
41 };
42
43 const getDataProfile = () => {
44   const entries = mockRepository.getMockData();
45   const streams = getAvailableStreamNames();
46
47   if (!entries || entries.length === 0) {
48     return null;
49   }
50
51   const profile = {
52     streams: streams,
53     entries: entries
54   };
55
56   return profile;
57 };
58
59 export { MockRepository, readProcessedData, getAvailableStreamNames, filterEntriesByStreamNames, toNumber, getDataProfile };

```

2) Correlations

Git Evidences



Video Evidence

Recap: Call with RACHIT GOYAL Saturday, | Meeting | Microsoft Teams

Recap: Leaders T1 2026 - IloTDM Thursday, May 7 | Meeting | Microsoft Teams

Code Walkthrough

<https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=01ebd428-a44a-4f51-ac1a-b445007665c9>

Code Demo

<https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=01ebd428-a44a-4f51-ac1a-b445007665c9>

Planner Governance Evidence (Software Architecture Bucket)

Planner board used for architecture task planning and progress traceability:

- <https://planner.cloud.microsoft/webui/plan/Zx8AozAXI0qGNjreoURensgADRuX/view/board?tid=d02378ec-1688-46d5-8540-1c28b5f470f6>

Planner Snippets

The screenshot displays the Microsoft Planner web interface. The browser address bar shows the URL: planner.cloud.microsoft/webui/mytasks/assignedtome/view/board?tid=d02378ec-1688-46d5-8540-1c28b5f470f6. The interface is titled 'DEAKIN UNIVERSITY Planner'. On the left, a sidebar contains navigation options: 'My Day', 'My Tasks' (selected), 'My plans', and 'Pinned'. The main area is divided into three columns: 'Not started', 'In progress', and 'Completed'. Each column contains task cards. The 'Not started' column has one card: 'ARCH-015: Prepare for implementation Phase'. The 'In progress' column has three cards: 'ARCH-013: Setup Release Notes Process', 'ARCH-014: Conduct Architecture Review', and 'ARCH-009: Document integration points'. The 'Completed' column has four cards, all marked as completed by 'ARJUN SINGH on...'. Each card includes a title, a description, a due date (15/05), and a progress indicator. The interface also features a search bar at the top and a 'New plan' button at the bottom left.

Certificate copy of the Course



Project Worklog

PROJECT_HOURS_LOG_WEEK1_9.docx

Research References Used

1. <https://www.linkedin.com/learning/software-architecture-foundations/the-importance-of-software-architecture?u=2104084>
2. <https://lms.simplilearn.com/courses/8356/Introduction-to-Software-Architecture>.
3. Deakin AI and Literature Review Guide
<https://deakin.libguides.com/ai-literature-review>